Geophys. J. Int. (2000) **142,** 241–256

Palaeomagnetism of Early Triassic limestones from the Huanan Block, South China: no evidence for separation between the Huanan and Yangtze blocks during the Early Mesozoic

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Accepted 2000 March 9. Received 2000 January 7; in original form 1999 January 29

SUMMARY

There has been extensive debate on the timing of the collision between the Huanan and Yangtze blocks ever since Hsu and co-workers (Hsu *et al.* 1987, 1988) proposed a Mesozoic collision model for South China tectonics. We report new rock and palaeomagnetic data from an Early Triassic limestone formation from the Huanan Block that help to constrain its tectonic history. The samples are from two localities [Jiupi (JT), and Huangben (LT)] in Lianzhou County, northern Guangdong province.

Only thermal demagnetization is effective in decomposing a multicomponent remanent magnetization. The majority of samples show three components: a low-temperature component (A), an intermediate-temperature component (B), and a high-temperature characteristic remanent magnetization (ChRM). The A component has a steep inclination and is probably drilling-induced. The B component clusters around the present geomagnetic field direction and is a Brunhes-age viscous remanence. Fold test results suggest that the JT ChRM is pre-tectonic, and that the LT ChRM is a syn- or pre-folding remagnetization. Samples from the two localities show quite similar acquisition and thermal demagnetization of the isothermal remanent magnetization (IRM), and suggest that magnetite carries the ChRM component. However, partial anhysteretic remanent magnetization (pARM) spectra, anisotropy of anhysteretic remanence (AAR), rock fabric, and morphology of magnetic extracts are quite different for the two localities. The AAR of JT samples shows low anisotropy (average 2 per cent), and reflects a composite compaction and strain fabric. The AAR of LT samples shows a high degree of anisotropy (average 10 per cent), and is clearly consistent with rock deformation fabrics. The magnetic extracts of LT samples are dominated by spherical, botryoidal Ti-poor magnetite, which is probably precipitated from tectonic fluids during deformation, while the JT magnetic extracts are dominated by rod-shaped Ti-poor magnetite, probably derived from the erosion of igneous rocks, although minor amounts of botryoidal magnetite and rod-like magnetite particles can be seen in JT and LT extracts, respectively.

The rock fabric and magnetic fabric data, and SEM observations of magnetic extracts corroborate the results of the fold test. The best clustered LT ChRM direction is $D = 294^\circ$, $I = -12^\circ$ ($\alpha_{95} = 12^\circ$); its pole position is 187°E, 19°S ($dp = 6^\circ$, $dm = 12^\circ$). The assumption of a reversed polarity for the JT ChRM direction ($D = 198^\circ$, $I = -20^\circ$, $\alpha_{95} = 9^\circ$) implies that the Huanan Block was in the northern hemisphere during the Early Triassic, and its pole position, 239°E, 68°N ($dp = 5^\circ$, $dm = 10^\circ$) is close to Early Triassic pole positions from the Yangtze Block. A normal polarity for the JT-site ChRM would require tectonic movements inconsistent with the local geology. Therefore, the Huanan and Yangtze blocks were not separated in the Early Triassic. The rotation of the LT locality may have been caused by the collision of the southeast China coastal terranes and/or subduction of the Pacific plate in the Late Mesozoic.

Key words: Huanan Block, magnetic fabric, palaeomagnetism, SEM, Triassic, Yangtze Block.

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INTRODUCTION

The South China Block is composed of three tectonic units (Fig. 1): the Yangtze para-platform (Yangtze Block), the South China fold belt (Huanan Block), and the southeast coast fold belt (Dongnanyanhai Block). The amalgamation of the Yangtze and Huanan blocks has long been interpreted as part of the Caledonian orogeny (e.g. Yang *et al.* 1986; Ren *et al.* 1987). However, Hsu and co-workers (Hsu 1980, 1987; Hsu *et al.* 1987, 1988, 1990) proposed a Mesozoic age for this collision based on the discovery of Mesozoic thrust structure in South China and the reinterpretation of the Late Precambrian metasedimentary rocks of the Banxi Group. This new interpretation of South China tectonics led to extensive debate about the age and the tectonic background of the Banxi Group, the existence of thrust windows, and the Huanan Alps model (e.g. Gupta 1989; Rodgers 1989; Rowley *et al.* 1989; Hsu *et al.* 1989a,b; Rodgers & Hsu 1990). Recently, Li *et al.* (1995) and Li, Z.X. (1998) proposed a new model for South China tectonics. They suggested that the Yangtze Block and the Cathaysia Block (including the Huanan Block and the Dongnanyanhai Block) were amalgamated in the late Proterozoic at around 1000 Ma, and are separated by a failed intracontinental rift activated during the break-up of Rodinia at around 700 Ma. Whether the Yangtze and Cathaysia blocks were separated by a wide ocean basin or a failed intracontinental rift in the Early Palaeozoic can be tested by Early Palaeozoic palaeomagnetic data, a task which is beyond the scope of this study. The purpose of this study is to use Early Triassic palaeomagnetic data to distinguish the Mesozoic collision model from the much older collision models.

According to the Mesozoic suture model of Hsu and co-workers, the collision between the Yangtze and Huanan blocks started in the Late Triassic (the Indosinian movement).

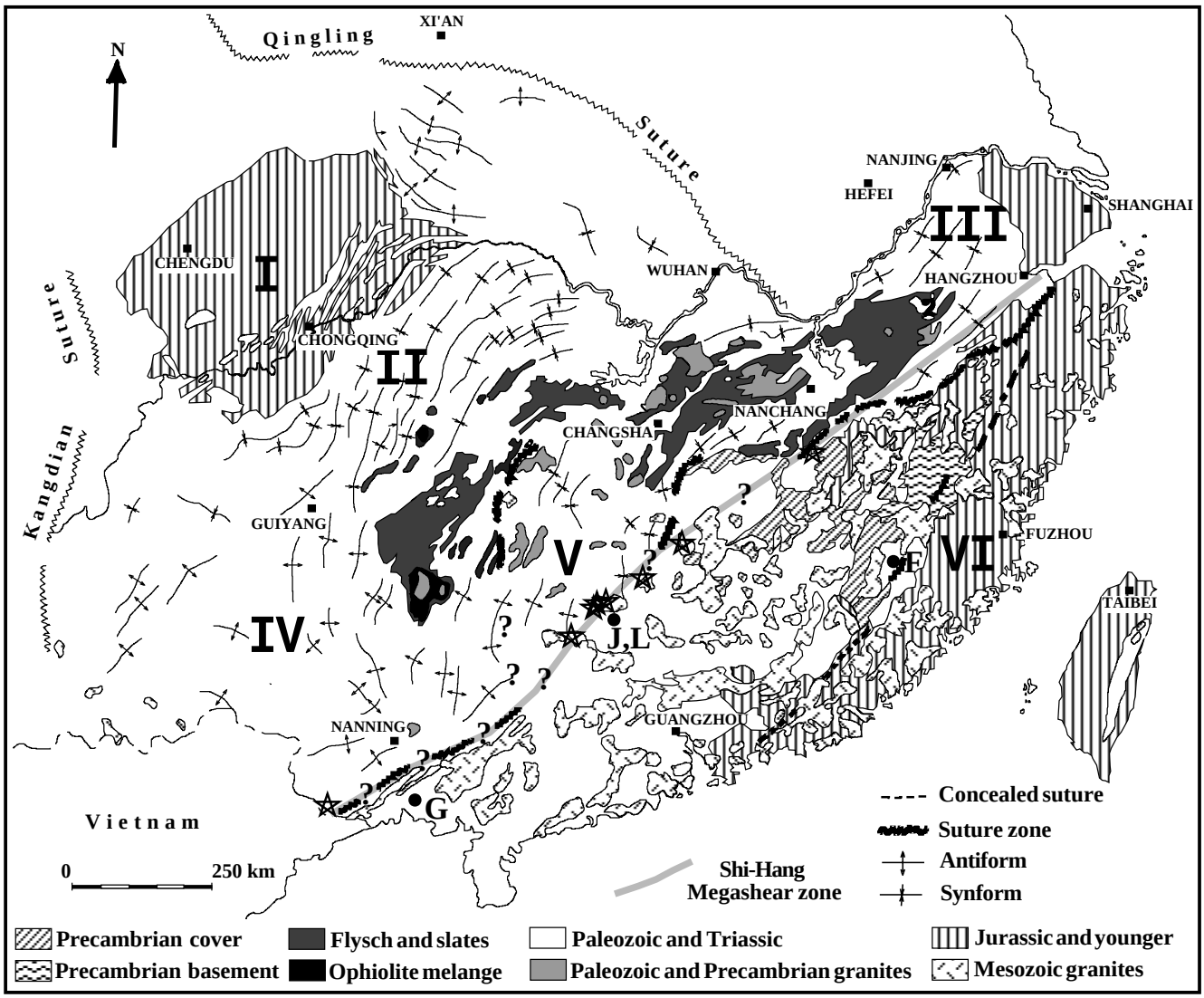


Figure 1. Geological sketch map of south China (after Hsu *et al.* 1988) showing the tectonic units: the Yangtze Block including the Sichuan Basin (I), Yangtze fold belt (II, III) and the western Guangxi fold belt (IV), the Huanan Block including the Huanan fold belt (V), and the southeast coast fold belt (VI). Zigzag lines and dashed lines indicate the possible suture and buried suture zones, respectively. The semi-transparent line is the Shi-Hang rift and megashear zone defined by Gilder *et al.* (1996). G and F are the localities studied by Gilder *et al.* (1995); J and L are the study localities reported in this paper; the locality studied by Chen *et al.* (1992) is somewhere near the J and L localities.

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One interpretation of the palaeomagnetic data is that the two blocks were separated latitudinally by a wide ocean basin in the Early Triassic (Chen *et al.* 1992; Zhai & Sequin 1991; Zhai *et al.* 1994; Gilder *et al.* 1995). However, as Gilder *et al.* (1995) point out, the Huanan Block could have been in either the northern or southern hemisphere, since the palaeomagnetic polarity has not been determined. Despite this ambiguity, the Huanan Block was reconstructed to lie in the southern hemisphere. Since, during the Early Mesozoic, the Yangtze Block was at low latitudes in the northern hemisphere, the polarity of the Early Triassic palaeomagnetic direction from the Huanan Block is extremely critical to a palaeomagnetic test of the Hsu tectonic model. Recent palaeomagnetic studies of Late Palaeozoic and Triassic limestones from the Yangtze Block indicate that the pre-folding characteristic remanent magnetization (ChRM) could be a remagnetization either of tectonic origin (Wang & Van der Voo 1993; Wang *et al.* 1993) or due to Cenozoic weathering (Huang & Opdyke 1996a). Rock magnetic studies become indispensable for understanding the origin and age of the Early Triassic limestone remanence and its tectonic implications for the Huanan block.

GEOLOGICAL SETTING AND SAMPLING

Our samples are from Lian County (24.2°N, 117.9°E) in the northern part of the Guangdong province, where the Daye Formation is representative of the Huanan region's lower Triassic littoral-neritic system (Yang *et al.* 1986; Guangdong Bureau of Geology & Mineral Resources (GDBGMR) 1988). The Daye Formation, which ranges in thickness from 170 to 800 m, conformably rests on upper Permian carbonate rocks and is overlain by middle Triassic intercalated sandstones and mudstones. The lower part of the Daye Formation is greyish thin-bedded limestone; the upper part is thick-bedded limestone. The boundary between middle Triassic and upper Triassic rocks is mostly unconformable and represents the Indosinian orogeny. The upper Triassic rocks are paralic to lacustrine sandy shales and coal seams, and intermontane molasses. The Daye Formation yields typical Early Triassic fossils, namely *Claraia wangi* and *C. Griesbachi* (pelycepod), and *Ophiceras sp.* (ammonoid) in the lower part, and *Eumorphotis cf. inaequicostata* and *Unionites fassaensis* (pelycepod), and *Meekoceras sp.* (ammonoid) in the upper part, which are equivalent to the Induan and Olenekian stages, respectively (Yang *et al.* 1986; Guangdong Bureau of Geology & Mineral Resources (GDBGMR) 1988).

The Huanan Block was clearly affected by the Indosinian orogeny (Late Triassic), as well as the Jurassic Yanshanian orogeny. The Indosinian orogeny caused folding of the Devonian to Triassic sedimentary cover accompanied by intrusion of granitic and granodioritic magma. The Yanshanian orogeny was characterized by widespread volcanic eruptions and intrusion of intermediate to acidic magma, forming the southeast China coastal volcanic rock belt and an extensive batholith (Ren *et al.* 1987; Guangdong Bureau of Geology & Mineral Resources (GDBGMR) 1988).

A total of 118 individually orientated samples were drilled from two localities with a gasoline-powered portable drill. Cores were orientated with an inclinometer and a magnetic compass. At locality Jiupi (JT) (Fig. 2), the Daye Formation is sporadically exposed along an 800 m long road-cut of the interstate highway (Route 107) extending from the Jiupi gas station to the Lianzhou City toll gate. The outcrop forms part of a monocline that dips gently towards the northeast. A total of 65 cores (eight sites) were collected from two exposures, which are less deformed, and cover about 50 m of section. The other locality (LT) is in a quarry near Huangben Village, close/to the Lianzhou–Shaoguan local road (Route 323), about 25 km east of the JT locality (Fig. 2). Deformation at LT locality is pervasive, as evidenced by fractures, cleavages, and calcite veins. About 40 m of thick-bedded limestone is exposed, dipping steeply towards the southeast. A total of 53 cores from seven sites were collected from this locality. At Zhejiang University, each core was cut into two or three 2.25 cm long cylinders with diameters of 2.49 cm. One sample from each core was mailed to Lehigh University for palaeo- and rock-magnetic studies.

PALAEOMAGNETISM

Remanent magnetization was measured by a CTF two-axis superconducting magnetometer. Four samples from each site were subjected to detailed thermal and alternating field (af) demagnetization, using a Schonstedt TSD-1 thermal demagnetizer and a Schonstedt GSD-5 af demagnetizer. A Sapphire SI-2 magnetic susceptibility meter was used to measure

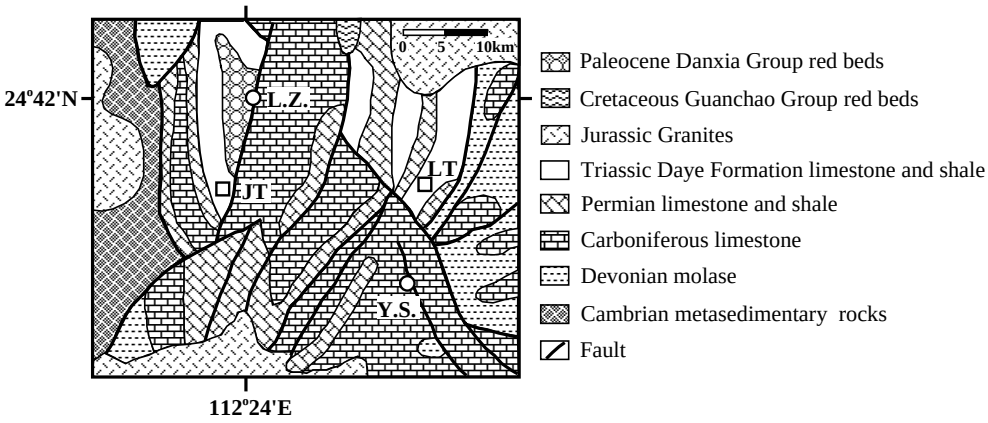


Figure 2. Geological map of Lianzhou area, northern Guangdong (from Guangdong Bureau of Geology & Mineral Resources (GDBGMR) 1988). Circles are the Lianzhou (LZ) and Yangshan (YS) cities. Squares indicate the sample localities of Jiupi (JT) and Huangben (LT).

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low-field room-temperature susceptibility after each thermal demagnetization step in order to monitor possible magnetic mineral alteration. The demagnetization data from each sample were analysed by principal component analysis (Kirschvink 1980) to resolve the components of magnetization. Fisher (1953) statistics were used to calculate the mean direction of each remanence component. Pilot thermal demagnetization (up to 575 °C) isolated three magnetization components (Figs 3 to 5, Table 1), while pilot af demagnetization was not able to resolve the multicomponent magnetization of the samples, although it did significantly (>95 per cent) decrease the intensity of natural remanent magnetization (NRM). Therefore, the remaining samples were all treated by progressive thermal demagnetization in 10 to 12 steps up to 575 °C.

The NRM directions of both the JT and the LT samples are steep with the LT NRMs showing better clustering (Figs 4a, 5a). NRM intensities of the JT and LT samples range from 0.4 to 0.7 mA m^{-1} and 4 to 10 mA m^{-1}, respectively. Component A had unblocking temperatures of less than 100 °C with an intensity of about 3/4 of the NRM intensity (Fig. 3). This low unblocking temperature component is distributed around a steep northerly direction (Figs 4b and 5b) and dominates the NRM direction. The A component may be drilling-induced, since there appears to be a correlation between core plunge and component A inclination (Fig. 6). A viscous origin of the A component caused by laboratory storage is unlikely, because the mean A component inclination in sample coordinates is much steeper than the field inclination of $\sim 58°$ in the storage space. In addition, some samples were stored with their Z-axes pointing up, while others were stored with their Z-axes pointing down. If a storage viscous remanent magnetization were important, some of the A component inclinations would be negative.

Component B was isolated at intermediate unblocking temperatures: 100–300 °C for JT and 200–300 °C for LT samples (Fig. 3). This intermediate unblocking temperature component is distributed around either the present dipole field (PDF) ($D = 0°$, $I = 42°$) or the present geomagnetic field (PGF) ($D = 357°$, $I = 35°$) directions (Figs 4c and 5c), and is presumably a Brunhes-age viscous remanent magnetization overprint. Component C was isolated between unblocking temperatures of 350 and 500 °C (Fig. 3). At higher temperatures, magnetic susceptibility increased significantly (Fig. 7), indicating the possible formation and/or alteration of magnetic minerals. This

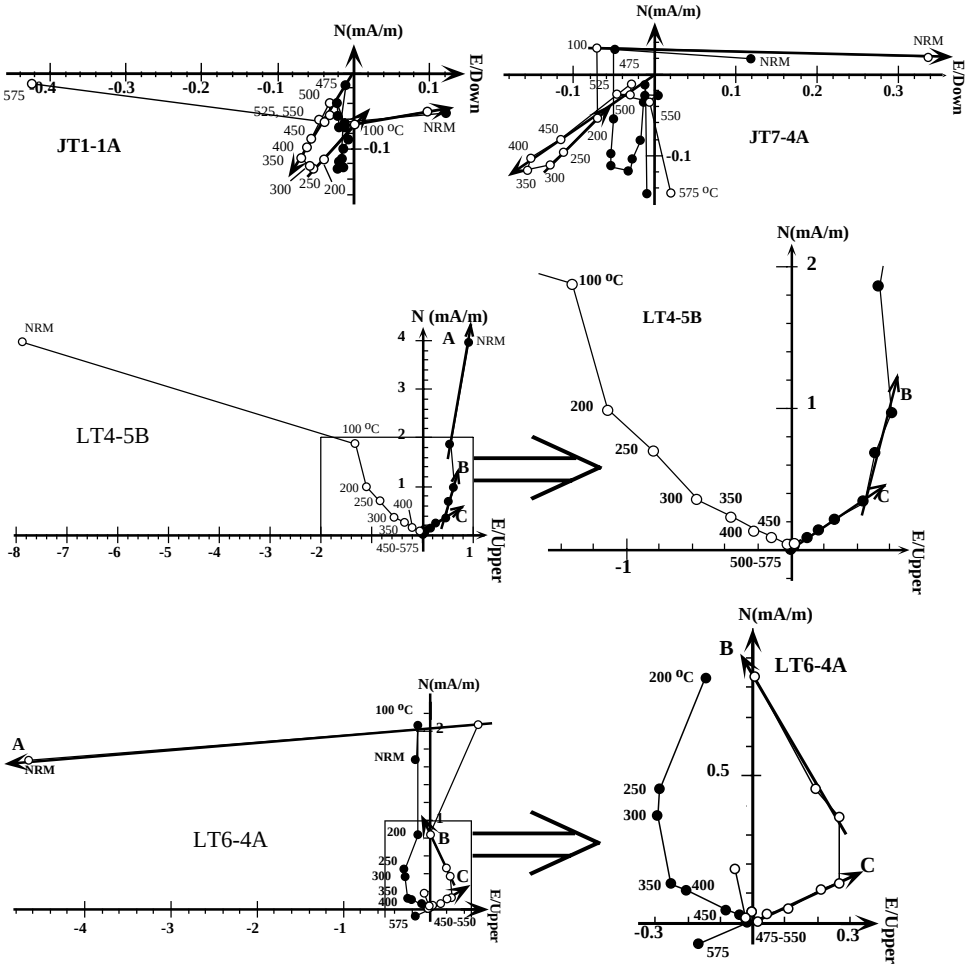


Figure 3. Vector demagnetization diagrams of typical thermal demagnetization results of the Triassic Daye Formation limestone samples from Jiupi (JT) and Huangben (LT) localities. Solid (open) circles are the horizontal (vertical) plane projections of remanent magnetization after progressive demagnetization indicated by temperature. The directions are in geographical coordinates. The thin arrows denote the three remanent components isolated by principal component analysis (Kirschvink 1980).

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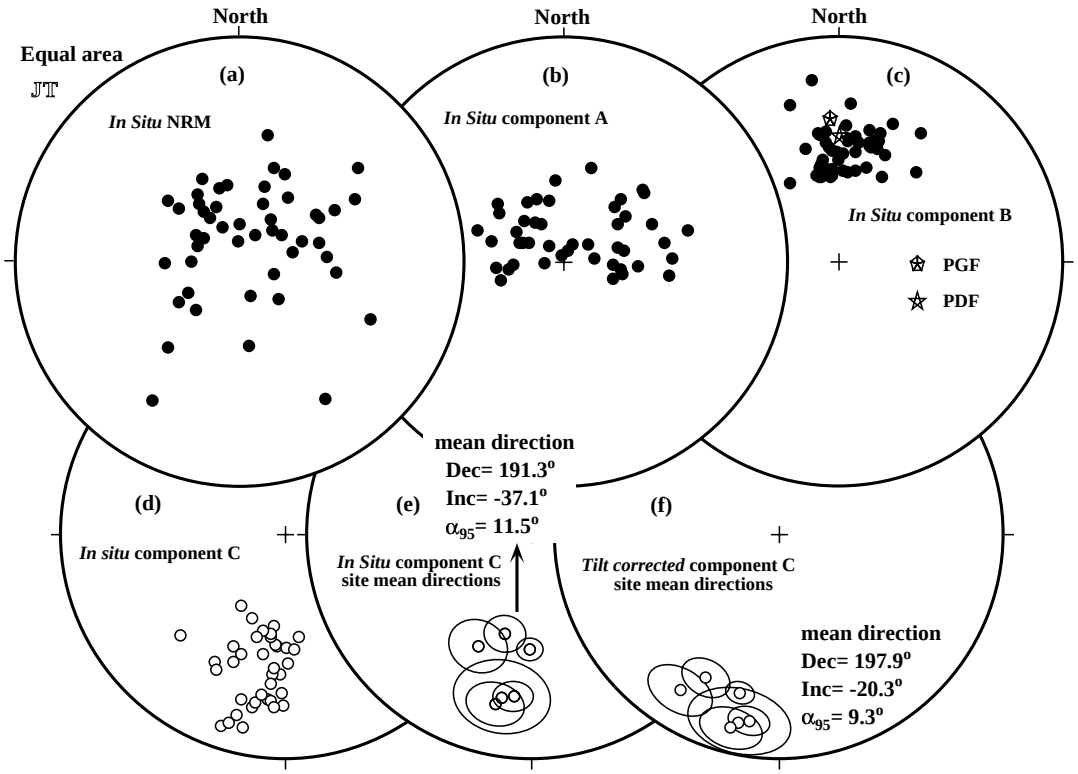


Figure 4. Equal-area stereonet projection of JT locality *in situ* sample NRMs (a), components A (b), B (c), and C (d), *in situ* site mean component C (e), and stratigraphic coordinates site mean component C (f). Solid (open) circles denote lower (upper) hemisphere projections.

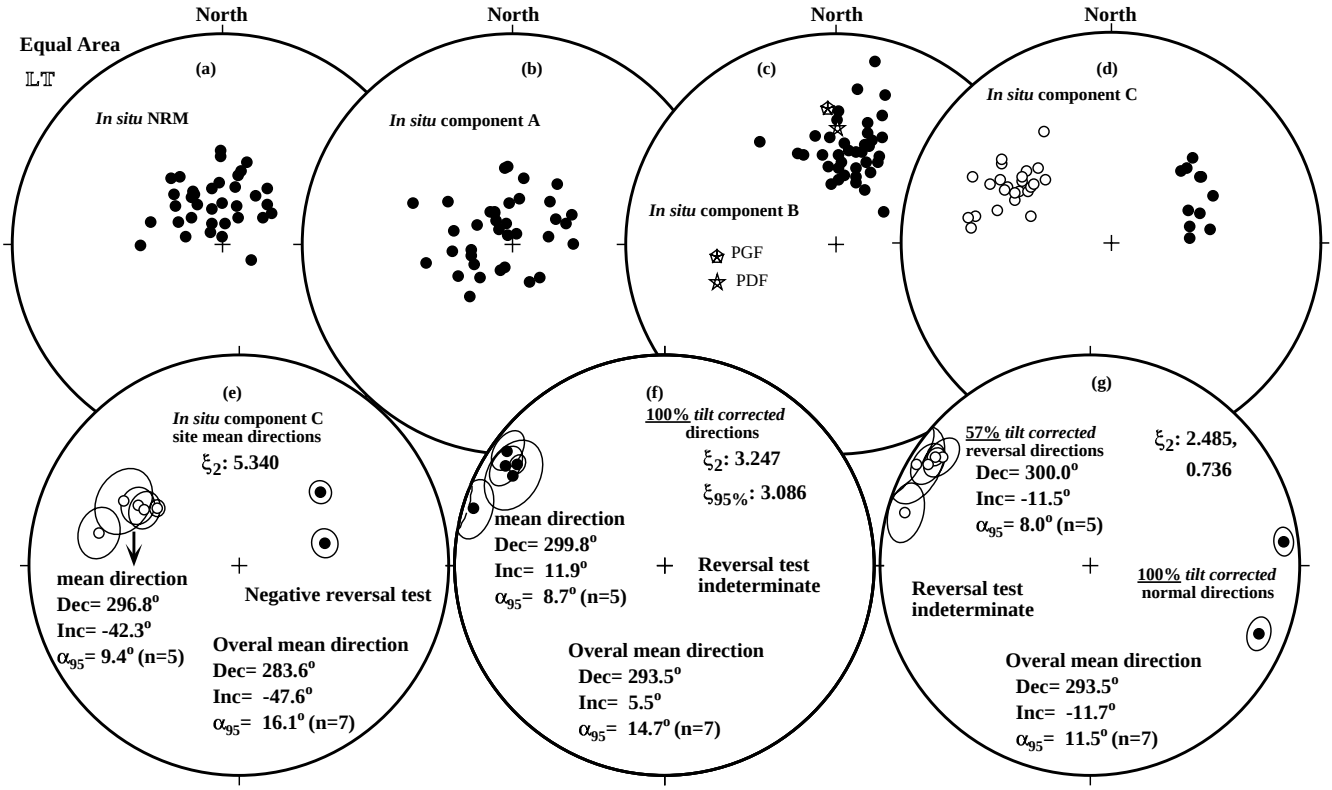


Figure 5. Equal-area stereonet projection of LT locality sample magnetic directions. Symbols are the same as in Fig. 4, except that the partial tilt-corrected reversal directions are plotted in (g).

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Table 1. Palaeomagnetic results of the Early Triassic limestones from the Huanan block, China.

Site	Comp.	*In situ*		Strike (°)	Dip (°)	Bedding		*n*/N	R	α_{95} (°)	κ
		Dec (°)	Inc (°)			Dec (°)	Inc (°)				
JT11	A	288.4	79.0	318.5	32.0	27.9	62.0	6/6	5.694	17.1	16.3
	B	5.4	50.9			20.2	24.5	6/6	5.852	11.7	33.9
JT10	A	307.1	67.3	320	30	5.8	57.4	6/7	5.931	7.9	72.2
	B	17.8	45.2			26.5	16.3	6/7	5.913	8.9	57.3
JT8	A	57.5	57.4	324	27	56.2	30.4	3/6	2.975	14.0	78.9
	B	12.3	45.8			23.6	23.7	6/6	5.965	5.7	141.5
	C	205.1	−43.8			212.3	−19.3	6/6	5.889	10.1	45.2
JT7	A	68.5	71.1	322	26	59.6	45.6	6/7	5.637	18.7	13.8
	B	1.9	52.1			18.1	32.4	7/7	6.900	7.8	60.3
	C	195.1	−51.8			206.8	−29.3	7/7	6.920	7.0	75.2
JT6	A	12.0	80.6	320	26	39.6	56.1	6/6	5.650	18.4	14.3
	B	355.6	49.5			11.9	31.2	6/6	5.860	11.4	35.7
	C	180.4	−47.6			194.4	−28.2	6/6	5.978	4.4	230.0
JT5	A	86.6	62.4	309	13	73.2	52.5	4/5	3.910	16.1	33.3
	B	7.9	41.5			12.4	30.1	5/5	4.877	13.6	32.5
	C	190.1	−26.7			192.4	−15.2	5/5	4.849	15.2	26.5
JT2	A	309.9	73.4	309	13	346.6	68.9	4/6	3.949	12.1	58.4
	B	353.2	51.1			2.2	41.3	6/6	5.977	4.5	218.1
	C	192.0	−23.7			193.8	−12.0	6/6	5.907	9.2	53.6
JT1	A	68.7	71.8	309	13	56.9	59.8	5/6	4.559	26.8	9.1
	B	358.6	51.9			6.8	41.4	6/6	5.928	8.1	69.8
	C	186.1	−27.9			188.9	−16.8	6/6	5.958	6.2	119.0
Mean C		191.3	−37.1					6/8	5.857	11.5	34.9
						197.9	−20.3	6/8	5.905	9.3	52.6
LT8	A	96.5	85.2	35	40	121.7	45.7	4/4	3.719	29.5	10.7
	B	10.0	56.1			69.9	51.9	3/4	2.978	13.1	89.0
	C	283.1	−31.9			−73.6	5.8	4/4	3.972	8.9	107.0
LT6	A	258.4	67.7	36	55	150.6	47.8	3/4	2.966	16.2	58.7
	B	4.0	44.3			66.3	45.3	4/4	3.841	21.7	18.9
	C	298.9	−37.4			−59.5	17.3	3/4	2.982	11.8	109.9
LT5	A	277.9	73.8	36	55	137.5	48.8	6/6	5.819	13.0	27.6
	B	357.0	46.8			69.3	50.4	6/6	5.708	16.6	17.1
	C	300.7	−43.4			−57.9	11.5	6/6	5.949	6.8	97.6
LT4	A	48.2	74.3	35	67	108.9	18.6	6/6	5.850	11.8	33.4
	B	15.4	52.7			83.3	29.9	6/6	5.934	7.7	76.1
	C	48.2	46.8			82.7	8.1	5/6	4.987	4.3	319.0
LT3	A	27.2	80.8	35	67	115.0	23.9	4/4	3.758	27.2	12.4
	B	20.4	54.3			85.7	26.9	4/4	3.857	20.5	21.0
	C	304.4	−50.8			−55.4	16.2	4/4	3.997	2.9	1008.9
LT2	A	351.4	83.8	49	57	134.8	38.2	6/6	5.793	13.9	24.1
	B	358.0	53.3			97.1	55.7	6/6	5.897	9.7	48.4
	C	300.1	−46.9			−53.9	8.3	6/6	5.953	6.6	105.5
LT1	A	303.4	75.5	56	60	153.6	43.2	7/7	6.683	14.2	18.9
	B	18.7	56.0			107.0	45.1	7/7	6.929	6.6	84.2
	C	75.4	54.9			112.0	14.1	7/7	6.956	5.2	135.2
Mean C*		296.8	−42.3					5/7	4.941	9.4	67.9
						299.9	11.9	5/7	4.949	8.7	77.8
		at 57 per cent tilt correction				300.0	−11.5	5/7	4.960	8.0	92.3
Mean C†						293.5	−11.7	7/7	6.788	11.5	28.3

* Sites LT1 and LT4 not included; † best clustered C component direction; Comp, magnetization component; Dec, declination; Inc, inclination; *n*/*N*, number of samples (sites) used to calculate mean direction versus number of samples (sites) demagnetized; *R*, length of the resultant vector; α_{95}, half angle of the 95 per cent confidence cone; κ, Fisher's precision parameter.

may explain significant intensity increases and randomization of the remanent magnetization after thermal demagnetization at the highest temperatures. Since component C decays towards the origin, it is the characteristic remanence (ChRM).

The JT ChRM is single polarity (Figs 4d to f), while the LT ChRM is dual polarity (Figs 5d to f). During stepwise unfolding, the Fisher precision parameter, *k*, for the JT ChRM increases from 35 in geographical coordinates to 53 in stratigraphic coordinates (Fig. 8a). The precision parameter for the LT ChRM increases from 15 in geographical coordinates to 18 in stratigraphic coordinates (Fig. 8b). Both the JT and LT ChRM directions fail to pass McElhinny's (1964) and McFadden & Jones's (1981) fold tests, probably due to the small variation of bedding attitudes within each of the outcrops

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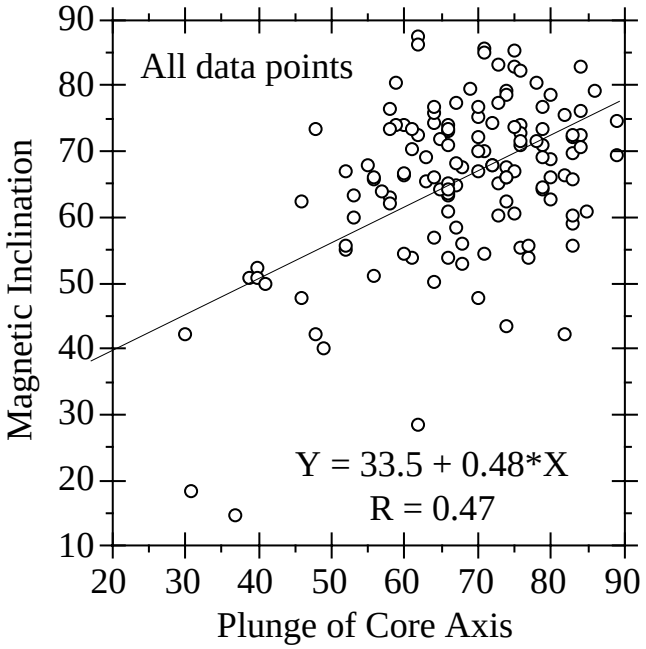


Figure 6. The correlation between the inclination of component A and the plunge of samples suggest that the low unblocking temperature overprint component is probably drilling-induced.

(Table 1). McFadden's (1990) fold test, which is based on the correlation between the remanence directions and their corresponding geometric coordinates, is more sensitive to bedding attitude variations. Because within each outcrop the dip directions fall in the same quadrant, the second definition, ξ_2, should be used to test correlation (McFadden 1990). Application of McFadden's (1990) fold test indicates that the JT ChRM direction in geographical coordinates is correlated with the present geographical coordinates at greater than 95 per cent confidence, while the bedding-corrected ChRM direction is not correlated with the stratigraphic coordinates. This suggests that the JT ChRM was acquired before tilting of the strata. For the LT ChRM, at 95 per cent significance, the *in situ* direction and tilt-corrected direction are not correlated with the geographical coordinates and stratigraphic coordinates, respectively, suggesting that the overall LT ChRM could be post-, syn-, or pre-tilting, and thus the fold test is indeterminate. Since ξ_2 in geographical coordinates is greater than ξ_2 in stratigraphic coordinates, there is a higher possibility that the LT ChRM was acquired at greater tilt correction.

A reversal test (McFadden & McElhinny 1990) for the *in situ* LT ChRM is negative, excluding the possibility of post-tilting remagnetization. However, a reversal test of the tilt-corrected LT ChRM is indeterminate because the two polarity directions both have low positive inclinations. Since the bedding-corrected inclinations of the two LT ChRM polarity groups all have shallow positive inclinations, one of the polarity groups must have been over-tilt-corrected. The positive *in situ* inclination ChRM group remains positive during progressive tilt-correction, while the negative *in situ* inclination ChRMs change into positive inclination at a high degree of tilt-correction. Progressive tilt-correction indicates that the precision parameter reaches a maximum at 57 per cent unfolding for just the five negative *in situ* inclination sites (Fig. 8c), although at 95 per cent confidence the fold test for these five sites is indeterminate. The best clustering of LT ChRMs occurs when the five negative *in situ* inclination sites are unfolded at 57 per cent and the two positive *in situ* inclination sites are corrected by 100 per cent unfolding. The statistical parameter ξ_2 was recalculated for all the LT ChRMs in geographical coordinates, stratigraphic coordinates, and the best clustered coordinates as defined above (Fig. 5). The results indicate that only the best clustered directions have no apparent correlation with their corresponding coordinates at 95 per cent significance, and thus pass McFadden's (1990) fold test. In summary, two palaeomagnetic directions were obtained (Table 1) from palaeomagnetic analysis: the pre-tilting, most probably Early Triassic direction (JT), $D = 197.9^\circ$, $I = -20.3^\circ$ ($\alpha_{95} = 9.3^\circ$), and the syn-tectonic, probably Late Triassic direction (LT), $D = 293.5^\circ$, $I = -11.7^\circ$ ($\alpha_{95} = 11.5^\circ$).

ROCK MAGNETISM

Severe Late Triassic to recent remagnetization of Palaeozoic and Early Triassic rocks in South China has been reported (e.g. Kent *et al.* 1987; Wang & Van der Voo 1993; Wang *et al.* 1993; Huang & Opdyke 1996a). In some cases, even those

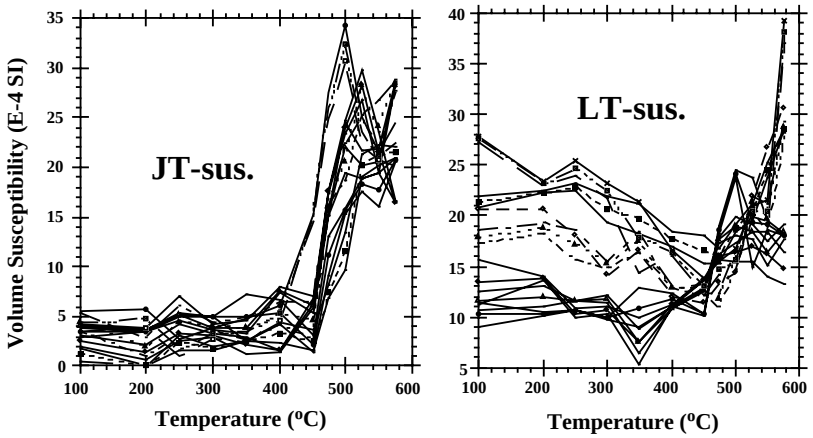


Figure 7. The variation of room-temperature magnetic susceptibility after thermal demagnetization indicates a rapid increase in susceptibility above 475 °C, which may cause scattering of remanence measurement.

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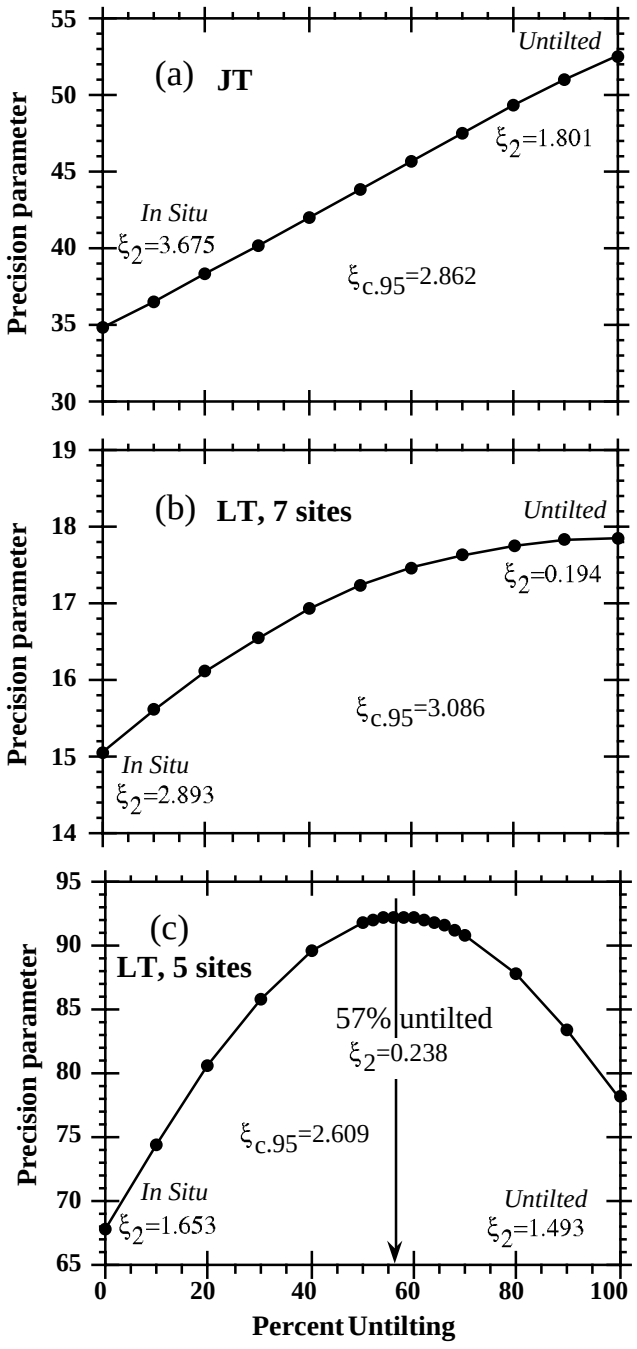


Figure 8. Fisher precision parameter (Fisher 1953) of mean JT (a) and LT (b,c) ChRM directions as a function of incremental tilt correction. The statistic parameters (ξ_2) of the fold test (McFadden 1990) are also shown. $\xi_{c.95}$ is the critical value at 95 per cent confidence. The test suggests that the JT ChRM is pre-folding, and that the age of the LT ChRM is indeterminate at 95 per cent significance. At lower confidence levels, it may suggest a pre- and syn-tilting age of the LT ChRM.

ChRMs that pass the fold test could have been remagnetized during tectonic events (Wang & Van der Voo 1993; Wang *et al.* 1993) or recent weathering (Huang & Opdyke 1996a), thus making the results of the fold test ambiguous. In fact, the JT *in situ* and tilt-corrected ChRM directions are very close to a reversed polarity B component or a reversed polarity PDF/PGF (Figs 4c and d, Table 1), and therefore the JT ChRM could be a remagnetization that occurred after tilting or even a recent remagnetization. However, comparison of ChRM directions with an overprint component does not provide sufficient information to determine the age of the ChRM. Rock-magnetic experiments, rock and magnetic fabrics, and scanning electron microscopy observations can be used to determine the magnetic mineralogy and the source and origin of the magnetic minerals, and hence the magnetization process for the JT and LT samples.

The acquisition of isothermal remanent magnetization (IRM) and back-field demagnetization were conducted on an ASC IM-10 impulse magnetizer. The acquisition and demagnetization curves show similar behaviour for the JT and LT samples (Fig. 9). In most cases, IRM increases rapidly at low fields (<0.2 T), and gradually increases until field intensities of 0.9 T are reached. All the samples are close to saturation at 0.9–1.2 T. This behaviour suggests the dominance of low-coercivity minerals such as magnetite and maghemite, and a minor amount of high-coercivity minerals, namely haematite, goethite, and pyrrhotite.

Thermal demagnetization of a high-coercivity (0.1–1.2 T) IRM component and an orthogonal low-coercivity (0–0.1 T) anhysteretic remanent magnetization (ARM) component acquired in a $\sim$50 μT DC field is also similar for JT and LT samples (Fig. 9). The ARM unblocking spectra are distributed between 100 and 550 °C, which is indicative of magnetite. The IRM decays more rapidly at low temperatures (<400 °C) than at high temperatures, and is completely demagnetized by 500–550 °C. At higher temperatures (>600 °C), IRM intensity drops below or to the magnetometer's noise level (1×10^{-11} A m^2), and thus haematite is unlikely to be a carrier of remanence. The unblocking of the IRM component at low temperatures (<400 °C) may be indicative of goethite, pyrrhotite, greigite, and/or inversion of maghemite to haematite. Unblocking of magnetite remanence may also contribute to the IRM decay at low temperatures. The inversion of maghemite to haematite might be ruled out by the lack of a significant decrease in magnetic susceptibility at temperatures above 400 °C (Fig. 7). Based on the rock-magnetic data, the carriers of the low- and intermediate-temperature components of magnetization could be goethite, pyrrhotite, greigite, or magnetite. Since the ChRM component is isolated between 350 and 500 °C, it is probably carried solely by magnetite.

The similar behaviour of JT and LT samples during IRM acquisition and thermal demagnetization could suggest that both JT and LT ChRMs are due to similar magnetization processes. However, the partial ARM (pARM) coercivity spectra, anisotropy of anhysteretic remanence (AAR), rock fabrics, and the morphology of magnetic extracts are quite distinct for the JT and LT samples. The LT pARM spectra peak at 10–20 mT and are dominated by low-coercivity particles, while the JT pARM spectra show a broad peak between 20 and 40 mT, and are skewed to low coercivities (Fig. 10).

Both JT and LT samples have experienced tectonic strain as evidenced by pressure solution seams and calcite veins. Thin stylolites and veins developed in JT samples, while strain in LT samples is more intensive with the development of pervasive solution seams. The stylolite planes of JT samples are close to the bedding plane except in three samples, while the vein planes are more scattered in orientation (Fig. 11a).

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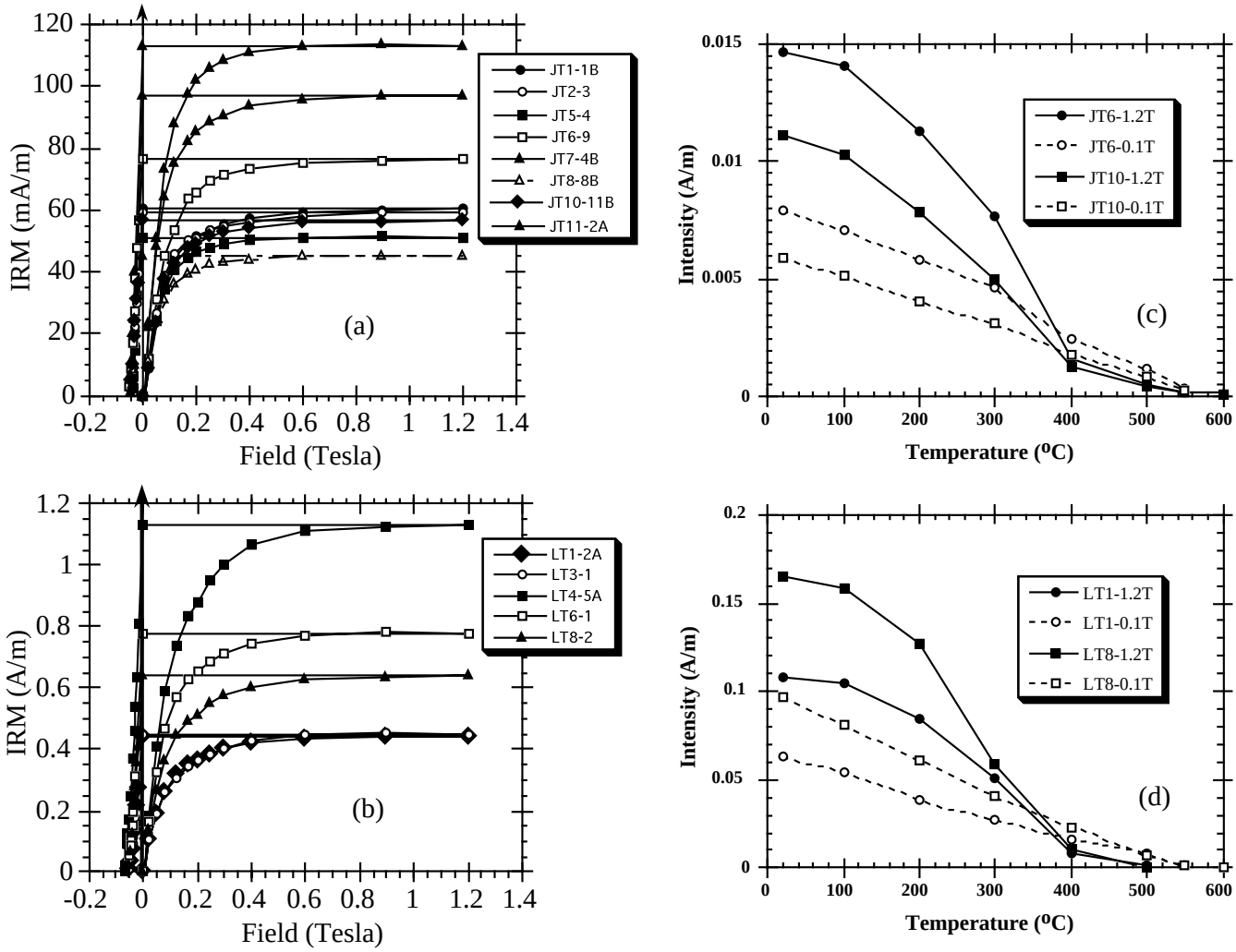


Figure 9. IRM acquisition and back-field demagnetization curves of JT (a) and LT (b) samples, and thermal demagnetization of orthogonal IRM and ARM components of JT (c) and LT (d) samples show a similar behaviour for JT and LT samples.

The relation between magnetic fabric and rock fabric may provide evidence for the origin of remanence. The AAR was measured for all the samples that were not thermally demagnetized. The ARM of JT samples was acquired in the coercivity window of 20–60 mT, while the ARM of LT samples was acquired in both low- (0–30 mT) and high- (40–80 mT) coercivity windows. JT samples show a low degree of AAR, ranging from 0.5 to 4 per cent (Fig. 11c), and scattered principal AAR axis directions (Fig. 11a). The minimum AAR axes are orientated near the pole to the bedding plane except for four samples in which the intermediate AAR axes are close to the pole to the bedding plane while the minimum axes are within the bedding plane. This is due to the low AAR degree of anisotropy. The JT samples apparently have a compaction fabric that has been affected by tectonic deformation. The low- and high-coercivity AARs of the LT samples are quite similar. The principal directions of LT AAR are well clustered, showing a triaxial fabric (Fig. 11b). The stylolite planes are subparallel to the bedding planes, which could suggest a compaction fabric; however, the minimum AAR axes lie within the stylolite planes rather than perpendicular to bedding planes, and thus the AAR data indicate a strong tectonic fabric rather than a compaction fabric. The maximum AAR axes lie within the vein planes and close to the intersection of the two sets of vein planes. Since the axis of maximum compressional stress must lie within the extension veins, the maximum axis of AAR fabric is apparently coincident with the maximum compressional stress direction, but not perpendicular to the maximum stress direction as would be expected. Since, for pre-existing magnetite grains, compressional deformation should lead to rotation of the minimum AAR axis parallel to the maximum stress direction, the coincidence between the maximum AAR axis and the maximum stress direction may suggest that the magnetite particles are not primary, but formed during or after deformation. The high degree of AAR (Fig. 11c), 6–16 per cent, is also indicative of a tectonic fabric. In summary, although both JT and LT rocks are deformed, the strain of the JT samples is low so that a compaction fabric still remains, as well as the primary remanence. For the LT samples, the initial compaction fabric has been completely replaced by a tectonic strain fabric, and it is reasonable to assume that the samples have been remagnetized during deformation, even though some of the magnetizations are pre-folding.

Scanning electron microscope (SEM) observation of magnetic extracts of JT and LT samples can also provide evidence for the origin of remanence. Limestone samples were dissolved

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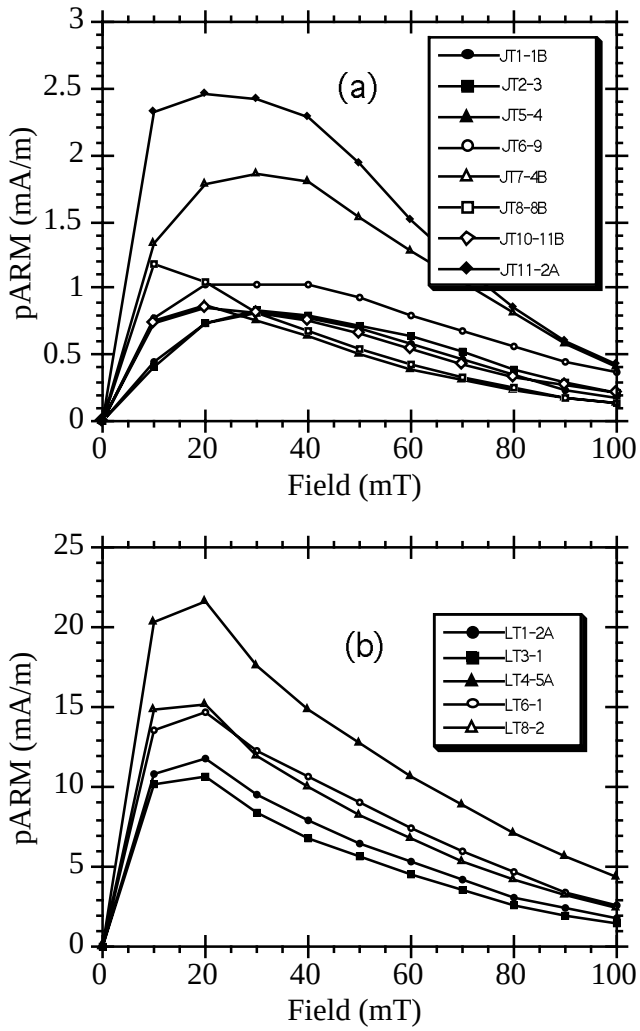


Figure 10. pARM spectra indicate different coercivity spectra for the JT (a) and LT (b) samples.

in buffered 2M acetic acid solution for 2 days. The residual material was rinsed in distilled water and the magnetic fraction was then extracted by a rare earth magnet. After air-drying, the magnetic extract was studied using a Jeol 6300F SEM.

The LT magnetic extracts are dominated by botryoidal spherical particles and a minor amount of rod-shaped particles (Fig. 12a). Energy-dispersive spectroscopy (EDS) (inset of Fig. 12a) indicates that the magnetic extracts are mainly of O and Fe elemental composition with minor amounts of Si and Ca. S and Ti are negligible. Botryoidal Ti-poor magnetite is commonly seen in remagnetized carbonate rocks (e.g. McCabe *et al.* 1983; Lu *et al.* 1990). The botryoidal shape (Fig. 12b) of magnetic particles observed in the LT extracts may suggest that they were precipitated from solution, probably during a tectonic event, similar to the case for particles in remagnetized North American carbonate rocks (McCabe & Elmore 1989).

The JT magnetic extracts are dominated by rod-shaped particles, and only a few botryoidal particles (Figs 12c and d). EDS of the JT magnetic extracts (inset of Fig. 12c) shows a large amount of O and Fe, moderate amounts of Si and S, and small amounts of Al, Mg, Na, and V and Cr (from the sample chamber). There is little or no titanium in the extracts, indicating that the magnetite is an almost pure iron oxide. EDS also suggests the presence of clay minerals and sulphide minerals, probably pyrrhotite or greigite. The rod-shaped particles are almost completely composed of O and Fe, although minimal amounts of Al, Si, and Ca are observed, probably due to attached clay minerals (Fig. 12d). The Ti-poor, rod-shaped magnetite is rarely seen in limestone samples, but it is common in igneous rocks in which magnetite–ilmenite intergrowths are formed by ilmenite exsolution from a magnetite–ultrospinel solid solution (e.g. Larson *et al.* 1969; Beske-Diehl 1988; Butler 1992; Hodych 1996) or magnetite crystallizing within the cracks of other mineral crystals, such as olivine or feldspar (Butler 1992; Zhang & Halls 1995; Warnock 1997). The rod-shaped magnetite is probably derived from the erosion of igneous rocks.

SEM observations suggest that the magnetic grain size is in the single-domain or pseudo-single-domain grain size range, consistent with the results of low-temperature demagnetization. Liquid nitrogen low-temperature demagnetization can only remove less than 10 per cent of NRM or IRM intensity, indicating that single-domain and/or pseudo-single-domain magnetite is the dominant magnetic carrier of ChRM (e.g. Levi & Merrill 1978; Dunlop & Argyle 1991; Hodych *et al.* 1998). Previous rock-magnetic and petrographic studies suggest that remagnetized carbonate rocks are dominated by low-coercivity particles of spherical secondary magnetite (e.g. McCabe *et al.* 1989). The pARM spectra of the LT samples are consistent with secondary, botryoidal magnetic particles as carriers of the remanence. The broad peak in the pARM spectra of the JT samples reflects the contamination of rod-shaped primary magnetite by botryoidal secondary magnetite as evidenced by the SEM observations. In addition, the LT intensities of NRM, ARM, IRM, and magnetic susceptibility are significantly higher (by about a factor of 10) than those of the JT samples. The magnetization process of LT samples is therefore more efficient than that for the JT samples. All these magnetic properties suggest that the LT ChRM was acquired during the precipitation of secondary magnetite particles from fluids during tectonic deformation, while the JT ChRM is probably of depositional origin. The LT ChRM has dual polarities, and therefore migration of fluids and magnetite precipitation may be multi-episodic, for example pre-folding and syn-folding, so that the acquired remanence may correspond to different stages of deformation.

DISCUSSION

The single-polarity C component of the JT samples is somewhat problematic, because it does not fit with the Early Triassic mixed-polarity geomagnetic field (e.g. Gradstein *et al.* 1995). Several other studies have also shown the predominance of a single reversed polarity in the Daye Formation from the Yangtze and the Huanan blocks (e.g. Gilder *et al.* 1995; Li *et al.* 1996; Tan & Kodama 1999). A normal polarity was observed in the mudstones from the bottom of the Daye Formation in the Yangtze Block (Li *et al.* 1996); however, the mudstones are not generally sampled because of their fragility during sample collection and preparation. Furthermore, these studies (Gilder *et al.* 1995; Li *et al.* 1996; Tan & Kodama 1999) were conducted mainly for tectonic purposes; therefore, the sites were not chosen to delineate the magnetostratigraphy.

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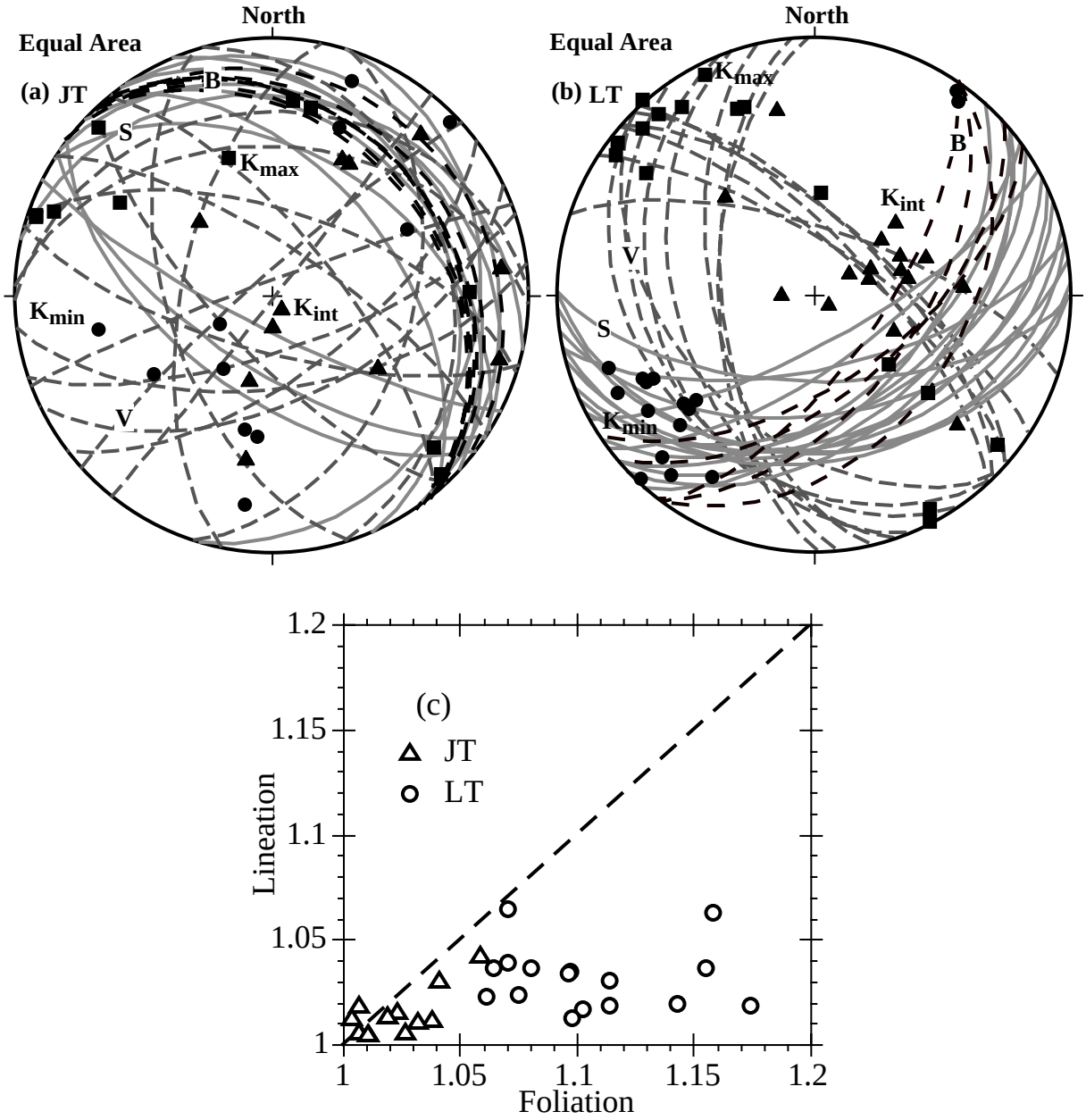


Figure 11. (a) and (b) are equal-area stereonet projections of *in situ* JT and LT sample rock fabric [stylolites (S) and veins (V)], and magnetic fabric (maximum, solid squares; intermediate, solid triangles; and minimum, solid circles, axes of AAR ellipsoid). Bedding planes (B) are also shown. (c) Plot of lineation (k_{max}/k_{int}) versus foliation (k_{int}/k_{min}) of magnetic fabrics of JT and LT samples. Figs 11–13 can be viewed in colour in the online version of the journal (www.blackwell-synergy.com).

The Daye Formation at the JT locality is typically exposed as cliffs due to the unique geomorphology of northwest Guangdong and Guangxi provinces. The distribution of sites is therefore often restricted to the lower part of the cliff exposures, and we were not able to sample a large stratigraphic section. In addition, a normal polarity direction in the JT samples, if it exists, is similar to the recent magnetic field direction (see Table 1 and Fig. 4), and it is difficult to isolate the normal polarity C component from the predominant Brunhes overprint component. Although it has been reported that chemical remagnetization of some carbonate rocks can be produced by burial diagenesis of clay minerals (i.e. smectite–illite conversion) (Katz *et al.* 1998), the magnetic extracts of the JT samples suggest a depositional origin for the ChRM (Figs 12c and d). The lack of dual polarities in our results may be simply due to our inability to sample a large enough section.

The ARM anisotropy of the LT samples shows an unexpected magnetic fabric, with its long ellipsoid axes at a high angle to bedding and lying in the vein sets (Fig. 11b). The coincidence of the long axes of the AAR ellipsoid with the intersection of the two vein sets, and a planar fabric parallel to one of the vein sets apparently suggest a tectonic origin for the magnetic fabric related to vein formation. Several other studies have tried to use AAR to study rock deformation and remagnetization processes (e.g. Jackson *et al.* 1989; Sun *et al.* 1993; Lu & Mccabe 1993), but the Daye Formation fabric does not conform with

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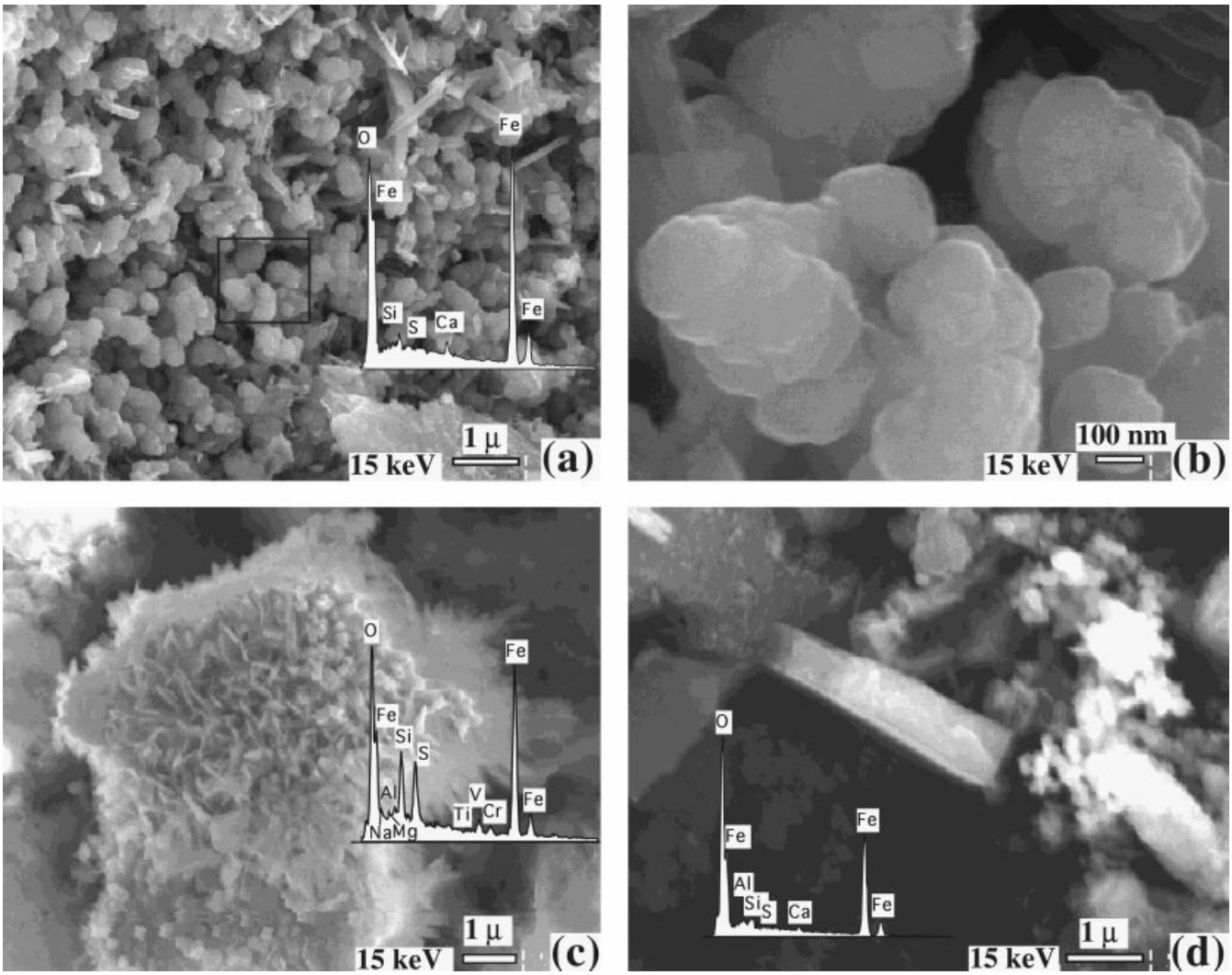


Figure 12. SEM micrographs and EDS spectra of magnetic extracts of LT (a,b) and JT (c,d) samples.

the results of these studies. The two vein sets cross-cut each other and have similar widths and lengths; therefore, they were formed at the same time. The spatial relationship between vein sets and magnetic fabric orientation might suggest that magnetic minerals were precipitated during vein formation. However, IRM acquisition results of subsamples from the veins suggest that there is almost no magnetite in the calcite veins. IRM data does, however, show a 40-times higher concentration of magnetic minerals in the stylolites than in the protoliths. Therefore, remagnetization and magnetic fabric formation were at an early stage of deformation before calcite vein precipitation.

In carbonates, pressure solution is an important deformation mechanism, in which magnetite may be concentrated along solution fronts and realigned by grain rotation (Jackson *et al.* 1989). In stylolites, the first-order tooth orientation (or stylolite columns) is interpreted to form parallel to the maximum principal compressive stress, which is oblique to the overall orientation of stylolite seams (e.g. Dunne & Hancock 1994). During pressure solution, magnetite may be mobilized and precipitated in stylolite seams. The observed magnetic fabric may be formed by precipitation or growth of magnetites, in pressure shadows. Those magnetite grains precipitated in the first-order teeth (columns) may tend to have their easy axes of magnetization rotated into the maximum principal stress direction. The spherical magnetite extracted from the LT samples and high magnetic concentration in the stylolites may also indicate that the AAR of LT samples is possibly a distribution anisotropy. However, an experiment conducted to test magnetic interaction of single-domain magnetites showed that the IRM acquisition and AF demagnetization curves intersect at $\sim$0.45 of saturation remanence, suggesting only a weak magnetic interaction between magnetite particles (Cisowski 1980) and probably indicating that a distribution anisotropy is not important. The most probable interpretation for the LT magnetic fabrics is the preferential alignment of secondary magnetites during pressure solution deformation.

The polarity of the ChRM from the lower Triassic Daye Formation is critical to our tectonic interpretation, because it determines whether the Huanan Block was in the northern hemisphere or the southern hemisphere during the time of magnetization. This will determine its position and tectonic movement (latitudinal transportation and vertical-axis rotation) with respect to the Yangtze Block, which was at low latitudes in the northern hemisphere during the Early Triassic. It is not possible to determine the Daye Formation's polarity by our

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study alone, since the Mesozoic apparent polar wander path for the Huanan Block has not been determined. We will instead argue for the Huanan Block's hemispherical position using the available palaeomagnetic data and geological observations.

Our results show that the LT ChRM is a syn-tectonic remagnetization. The LT direction and its pole position are very close to results reported by previous studies of units from the northern Guangdong (Chen *et al.* 1992) and Fujian provinces (Gilder *et al.* 1995) (Fig. 13). Since few rock-magnetic data, which would suggest the origin of the magnetic minerals, have been reported in these studies, it is difficult to assess the origin and age of their ChRMs. Although Chen *et al.* (1992) conducted IRM acquisition, IRM demagnetization, and low-temperature experiments to identify magnetic mineralogy, these experiments did not indicate whether the ChRMs were primary or secondary. In addition, Chen *et al.*'s (1992) data were not constrained by a standard fold test. The ChRM's of Gilder *et al.* (1996) from Fujian pass the fold test, yet a syn-folding (early stage of deformation) or pre-folding remagnetization cannot be ruled out because a positive fold test only suggests a pre-folding age for the ChRMs. Furthermore, the fact that the maximum AAR axis has a similar declination to the ChRM suggests that the ChRM may be realigned by grain-scale strain. Mesozoic deformation is pervasive in the Huanan Block and probably more intense in southeast China. Therefore, the previously reported ChRM directions may also have been affected by deformation.

Since the JT ChRM appears to be a primary magnetization, we will use it to argue for the polarity of the Daye Formation.

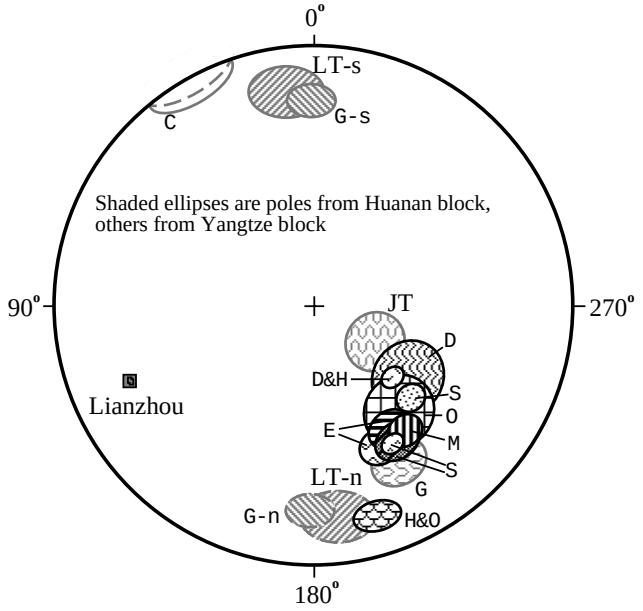


Figure 13. Triassic palaeomagnetic north pole positions from the Yangtze and Huanan blocks. The Yangtze Block poles are from Dobson *et al.* (1993) (D), Dobson & Heller (1993) (D&H), Enkin *et al.* (1992) (E), Huang & Opdyke (1996b) (H&O), Ma *et al.* (1993) (M), Opdyke *et al.* (1986) (O) and Steiner *et al.* (1989) (S). The Huanan Block poles are from Chen *et al.* (1992) (C), Gilder *et al.* (1995) (G), and this study (JT and LT). G, G-n, JT, LT-n are the north poles assuming a northern palaeolatitude of the block, and C, G-s, LT-s are the north poles assuming a southern palaeolatitude. Dashed ellipses and arc indicate southern hemisphere projection of the poles.

Assuming that the JT ChRM is representative of the Early Triassic palaeomagnetic field of the Huanan Block, if the Block were in the southern hemisphere during the Early Triassic the polarity of the ChRM would be normal. This would require a complicated kinematic history: both northward translation of more than 2000 km and either clockwise rotation of about 198° or counterclockwise rotation of about 162° since the Early Triassic. In contrast, if the Huanan Block were in the northern hemisphere during the Early Triassic, the polarity of the JT ChRM would be reversed and no significant latitudinal offset and dramatic Block rotation would be required. The Early Triassic pole position (JT) from the Huanan Block, assuming a reversed polarity ChRM, is close to the poles from the Yangtze Block of the same age (Fig. 13). This also suggests that these two blocks were already joined together in the Early Triassic. The small angular difference between the Huanan and Yangtze palaeopoles could be due to intracontinental deformation, which was apparently common within the Yangtze Block as indicated by the scatter of Yangtze pole positions (Fig. 13).

A normal polarity choice for the Daye Formation ChRM would require the Huanan Block to be a far-travelled terrane from the southern hemisphere. The Huanan Block should therefore display a distinctive pre-Triassic geological history, but it does not. Although there are some regional variations in the depositional facies and timing of volcanism in both the Yangtze and Huanan blocks, the general sedimentation pattern for the Devonian–Middle Triassic systems is very similar for the two blocks, especially the Carboniferous–Early Permian carbonates, the Late Permian continental–littoral coal measures, and the Lower Triassic littoral–neritic carbonates (Yang *et al.* 1986; Ren *et al.* 1987).

Gilder *et al.* (1996) used Sm–Nd and Rb–Sr isotopic compositions to define crustal provinces in South China. Their data suggest a homogeneous Proterozoic crust with no suture occurring in the Mesozoic. According to Hsu *et al.*'s (1988) model, the Yangtze Block subducted under and was accreted to the Huanan Block during the Late Triassic, with a wide ocean basin between the two blocks before the collision. However, there is no geological evidence of volcanic activity or magmatic intrusions during the Late Palaeozoic and Early Mesozoic in the middle and eastern Yangtze Block, adjacent to the Huanan Block. Had the oceanic crust between these two blocks existed, its subduction beneath the Huanan Block should have left a signature of that event, including deformation and magmatism. Geological evidence indicates that magmatic intrusion and volcanism in the Huanan Block started in the Late Triassic. Deformation of continental cover rocks was initiated at almost the same time (Late Triassic) in both the Yangtze and Huanan blocks. The stratigraphic records and tectonic histories of the Yangtze and Huanan blocks do not support large-scale relative displacement of the Huanan Block during the Late Palaeozoic and Mesozoic.

Gilder *et al.* (1996) defined a rift and megashear zone (Fig. 1) and a tectonostratigraphic terrane based on a Sm–Nd and Rb–Sr isotopic anomaly. The terrane includes the southeast China coastal belt and part of the Huanan Block, including northern Guangdong. If this terrane travelled from 2000 km further south along the megashear during the Mesozoic, it would require about 180° of vertical-axis rotation. If the declinations of our LT ChRM and other data (Chen *et al.* 1992; Gilder *et al.* 1995) are representative of the whole Block,

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and the JT site has been affected by local rotations, about 80° clockwise rotation, or 100° counterclockwise rotation, and more than 2000 km northward transportation would be required in the Mesozoic. However, the large-scale rotation of the Huanan Block would palaeogeographically reconstruct the terrane such that what was initially the southern end is now the northern boundary, and bring about a dramatic stratigraphic and isotope composition discontinuity. This would also cause one side of the Shi-Hang rift zone of Gilder *et al.* (1996) to be far from the present location of the rift zone. There is a general southeast younging of granitic intrusions in the Huanan Block (Wang *et al.* 1976; Li, X.H. 1998), which is interpreted to indicate the accretion of coastal terranes (e.g. Guo *et al.* 1984; Charvet *et al.* 1996) and the migration of the Pacific plate subduction boundary (e.g. Jahn *et al.* 1976; Chen 1992). The far-travel and significant vertical-axis rotation of the Huanan Block would result in a complicated distribution pattern for the timing of granitic intrusion. A large-scale Block transportation and rotation model seems difficult to fit with the observed geology. The LT ChRM therefore only indicates local vertical-axis rotations.

A reversed polarity ChRM for the JT magnetization provides a more reasonable explanation for the coherent depositional history for the Yangtze and Huanan blocks during the Late Palaeozoic and Early Mesozoic. It would require about 20° of vertical-axis rotation and latitudinal transport would not be necessary. The Early and Middle Triassic poles from the Yangtze Block lie along a small circle around the South China Block (Fig. 13). This is probably due to relative rotations between various sampling localities within the block. The dominant declination of the Early Triassic Yangtze Block is between 35° and 50°. The declination of the JT ChRM is about 20°. Therefore, the Yangtze Block has rotated clockwise by 15° to 30° with respect to the Huanan Block since the Late Triassic. This rotation may be responsible for the formation of rift basins in the Huanan Block, especially the Qingpang Aulogen and the Shi-Hang rift zone as defined by Gilder *et al.* (1996).

South China has experienced multiple tectonic events since the Triassic (e.g. Ren *et al.* 1987). Our palaeomagnetic and rock and magnetic fabric data suggest at least two episodes of deformation. The first deformation caused the remagnetization and tilting of the Daye Formation at the LT locality. The rock fabric (Fig. 11b) suggests a maximum palaeocompressional stress in a NW–SE direction. The palaeomagnetic declination is about 120° (using a normal polarity direction), so after correcting for the palaeomagnetically determined vertical-axis rotation, the major compressive palaeostress axis was approximately in the N–S direction during deformation. This stress is consistent with an episode of N–S convergence between the North China and South China blocks and may suggest that the Huanan Block is a coherent part of the South China block. The second event is the 80°–100° clockwise rotation of the LT locality, and probably other areas, for example Fujian (Gilder *et al.* 1995), with respect to the JT locality and the Yangtze Block. Southeast China is dominated by NE–SW-trending sinistral strike-slip faults, which were active in the Late Mesozoic and Cenozoic (e.g. Xu *et al.* 1987). Strike-slip faulting may have accommodated the observed Block rotation. This event may also be related to the accretion of the southeast China coastal terranes or the subduction of the Pacific Plate during the Mesozoic. Although the declination difference between the JT and LT sites could be explained by grain-scale strain, it may also be explained by local faulting. The area surrounding the JT and LT sites is complexly deformed and contains many faults (Fig. 2); hence, local Block rotations can probably explain the approximately 90° relative rotation between the sites.

Basically, three perplexing geological problems in South China led Hsu and co-workers to propose a Mesozoic collision model for the Yangtze and Huanan blocks. The problematic age of the Banxi Group, Mesozoic thrust structures in the Huanan fold belt, and a Mesozoic ophiolite melange do not, however, necessarily contradict a Caledonian-age amalgamation between the Yangtze and Huanan blocks. Recent work suggests that both suites of rocks in the Banxi Group, a sedimentary cover sequence, and an older metamorphic complex that includes ophiolite blocks are pre-Sinian in age rather than Mesozoic, based on stratigraphic relationships, isotopic dating, and palaeontological data (Liu *et al.* 1996). Only one of the Mesozoic tectonic windows has been confirmed by subsequent geological mapping—the Xishan thrust structure window in Jiangxi province (Liu *et al.* 1996). This result does not support Hsu and co-workers' prediction that more tectonic windows would be found to support their Mesozoic collision model. The Xishan window is a southward-verging Mesozoic overthrust structure that can be easily explained as intracontinental deformation due to the indentation of the North China Block into the South China Block during convergence of the South and North China blocks (Tan & Kodama 1999). Finally, ophiolite melange is not common in the South China block; however, a subduction complex has been observed in Jiangxi, Zhejiang, and Fujian provinces, indicating a continental collision zone (e.g. Wang *et al.* 1988; Zhejiang Bureau of Geology & Mineral Resources (ZBGMR) 1988; Gao *et al.* 1993). This is a Late Mesozoic collision zone which is probably due to the accretion of at least part of the southeast China coastal terrane, rather than the amalgamation between the Yangtze and Huanan blocks. It seems that all the geological features that led Hsu and co-workers to postulate a Mesozoic collision are consistent with a Caledonian collision model for the Yangtze and Huanan blocks.

CONCLUSIONS

This study yields pre-tectonic and syn-tectonic palaeomagnetic directions. Although IRM acquisition and thermal demagnetization curves are similar in both remagnetized and un-remagnetized samples, pARM spectra, magnetic and rock fabrics, and magnetic extracts are quite different for LT and JT, which suggests a depositional origin and secondary chemical origin of remanence for the JT and LT sites, respectively. Rock-magnetic data are indispensable for constraining the age, source, and mechanism of magnetization. Our data suggest that the Huanan Block was already part of the South China Block in the Early Triassic. Some areas in the Huanan Block may have been affected by Mesozoic accretion of the southeast coastal terranes and/or the subduction of the Pacific plate as evidenced by vertical-axis rotation. The Indosinian deformation in the Huanan fold belt is probably within the same tectonic domain as the collision between the North and South China blocks.

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ACKNOWLEDGMENTS

We are indebted to Mr Wu Nengyou at the Guangzhou Marine Geology Bureau, the Ministry of Geology and Mineral Resources, for his knowledge of local geology. We thank Jason Ali and an anonymous reviewer for their critical comments. This study was part of a youth grant awarded to XT by the National Natural Science Foundation of China (NNSFC) (49404049). Field work of this study was supported by a NNSFC grant (49574216) to DF. Laboratory work was supported by a NSF grant (EAR-9315778) to KPK.

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